

Seth DeWhitt

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EDUCATION

Purdue University, *West Lafayette, IN*

Aug. 2023 – May 2027

B.S. in Computer Science, Software Engineering concentration

GPA: 3.6/4.0

Relevant Coursework: Data Structures, Analysis of Algorithms, Software Engineering, Systems Programming, Computer Architecture, C Programming, Software Testing, Object-Oriented Programming, Linear Algebra

Activities: Purdue Jazz Education Network (President), Purdue Jazz Band (Lead Tenor Sax.)

Awards: Elmhurst University Jazz Festival Outstanding Soloist (2x), Dean's List

EXPERIENCE

The Data Mine, Purdue University | *Git, Agile, Python, Typescript*

Aug. 2025 – Present

Project Lead & Teaching Assistant

West Lafayette, IN

- Directing a team of 11 students in collaboration with the Practical AI Podcast (PAI) and Prediction Guard
- Managing implementation of a RAG chatbot with vector search to link responses to episode transcripts/timestamps
- Leading design of full-stack podcast generation app using user-uploads and voice clone models trained on PAI hosts
- Planning/Leading biweekly Agile sprints, coordinating tasks and progress reviews across a 50k+ line codebase

Chewy | *Python, Typescript, FastAPI, Docker, AWS, Terraform, Jenkins, PostgreSQL*

June 2025 – Aug. 2025

Software Engineering Intern

Bellevue, WA

- Streamlined data science workflows via provisioning dev environments on AWS ECS loaded with custom tooling
- Decreased admin overhead by **50%** and enhanced service scalability using AWS Fargate
- Enabled secure persistent file storage across dev sessions by linking AWS EFS to ECS and an RDS database
- Automated ECS, EFS, & database maintenance with AWS Lambda and Kinesis, lowering AWS expenses by **25%**
- Built application and AWS IaC using Terraform and a Jenkins CI/CD pipeline, streamlining deployment

The Data Mine, Purdue University | *Python, R, Sci-kit learn*

Aug. 2023 – May 2025

Undergraduate Data Science Researcher

West Lafayette, IN

- Enabled climate change adaptation for Bayer farming clients by modeling future impact on corn yields (2023-2024)
- Optimized a LightGBM model for genomic datasets, improving Pearson correlation and reducing RMSE by **24%**
- Integrated machine learning models with GxExM to predict product placement for Syngenta (2024-2025)
- Implemented data preprocessing pipelines for quality control via mixed linear models & filtering genetic markers

PROJECTS

MyPort | *TypeScript, React, Next.js, OpenAI, LangChain, Supabase* | [GitHub](#) | [Link](#)

Oct. 2025

- Pioneered RAG chatbot to answer portfolio questions via Supabase, OpenAI embeddings, and LangChain retrievers
- Built automated ingestion pipeline to chunk portfolio data and autosync, supporting idempotent updates/backfills

Evallm | *TypeScript, React, Next.js, Vercel, OAuth2, MongoDB* | [GitHub](#) | [Link](#)

July 2025 – Aug. 2025

- Developed full-stack app deployed on Vercel using Groq API to evaluate user-prompted Meta LLM responses
- Implemented analytics dashboard with prompt analysis, LLM stats, and experiment tracking for model comparison
- Built responsive UI with real-time insights and Google authentication to enhance user experience

UNIX-like Shell | *C++, Lex/Yacc*

Feb. 2025 – Apr. 2025

- Developed a feature-rich command-line shell in C++ using Lex/Yacc for advanced parsing and execution.
- Implemented I/O redirection, pipes, subshells, signal handling (Ctrl-C, zombie elimination), and built-in commands
- Enhanced user experience with environment variable expansion, command history, and other interactive features

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, Java, C/C++, SQL, R

Cloud/Data: AWS, boto3, pycopg2, PostgreSQL, NoSQL, Pandas, Flyway, Supabase

Dev Tools: Git, FastAPI, React, Node.js, Next.js, Vite, Tailwind, Docker, Pinecone, MongoDB, Terraform, Jenkins